

A PROJECT TITLE
ON
CROP YIELD PREDICTION USING RANDOM FOREST, XGBOOST, AND
SUPPORT VECTOR REGRESSION TECHNIQUES

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Crop Yield Prediction Using Random Forest, XGBoost, and Support Vector Regression Techniques

1. Abstract:

Agricultural productivity forecasting plays a vital role in ensuring food security, optimizing resource allocation, and supporting informed decision-making for farmers and policymakers. Traditional statistical approaches to crop yield estimation often fail to capture the complex, non-linear relationships between yield and factors such as rainfall, temperature, soil composition, and fertilizer usage. This project proposes a machine learning based framework for crop yield prediction that leverages three regression techniques: Random Forest, XGBoost, and Support Vector Regression (SVR).

The dataset used comprises historical agricultural records including weather parameters, soil characteristics, and past yield values across multiple regions and crop varieties. After preprocessing steps such as handling missing values, encoding categorical variables, and feature scaling, the dataset is split into training and testing subsets. Random Forest, an ensemble method based on bagging of decision trees, is employed for its robustness to overfitting and ability to handle high-dimensional data. XGBoost, a gradient boosting framework, is used for its predictive accuracy and efficient handling of structured tabular data through regularization. Support Vector Regression is applied using kernel-based transformations to capture non-linear dependencies between input features and target yield.

The performance of the three models is evaluated using regression metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R-squared). Hyperparameter tuning is carried out using grid search and cross-validation. Experimental results demonstrate that ensemble-based methods, particularly XGBoost, achieve higher prediction accuracy compared to SVR, while Random Forest offers a strong balance between interpretability and performance. Feature importance analysis further reveals that rainfall, soil nutrient levels, and temperature are among the most significant predictors of crop yield.

This comparative study offers insights into the strengths and limitations of different machine learning algorithms for agricultural yield forecasting, helping stakeholders make data-driven decisions regarding crop planning, resource management, and risk mitigation.

2. INTRODUCTION:

Agriculture remains the backbone of many economies worldwide, and accurate prediction of crop yield is essential for effective planning at both the farm and policy levels. Yield is influenced by a wide range of interdependent factors, including climatic conditions, soil quality, irrigation practices, and the use of fertilizers and pesticides. The inherent complexity and non-linearity of these relationships make conventional statistical models inadequate for capturing the true dynamics of crop production.

With the advancement of machine learning, data-driven approaches have emerged as powerful tools for modeling such complex agricultural systems. Machine learning algorithms can learn patterns directly from historical data without requiring explicit assumptions about the underlying relationships between variables. Among the various algorithms available, ensemble methods such as Random Forest and XGBoost, along

with kernel-based methods like Support Vector Regression, have shown considerable promise in regression tasks involving structured, tabular data.

This project aims to design and implement a crop yield prediction system that compares the performance of Random Forest, XGBoost, and Support Vector Regression models. By analyzing historical data on weather, soil, and crop parameters, the system predicts crop yield for a given set of input conditions. The comparative evaluation of these three algorithms helps identify the most suitable model for this domain, considering both predictive accuracy and computational efficiency. The insights generated from this project can support farmers and agricultural planners in making proactive, data-informed decisions to improve productivity and reduce losses due to unforeseen environmental variability.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10 / 11 or Linux (Ubuntu 20.04+)
- Programming Language: Python 3.9 or above
- IDE: Jupyter Notebook / Google Colab / VS Code
- Libraries: Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn
- Version Control: Git and GitHub
- Hardware Requirements: Minimum 8 GB RAM, Intel i5 processor or equivalent

Signature of the Guide

Course Instructor